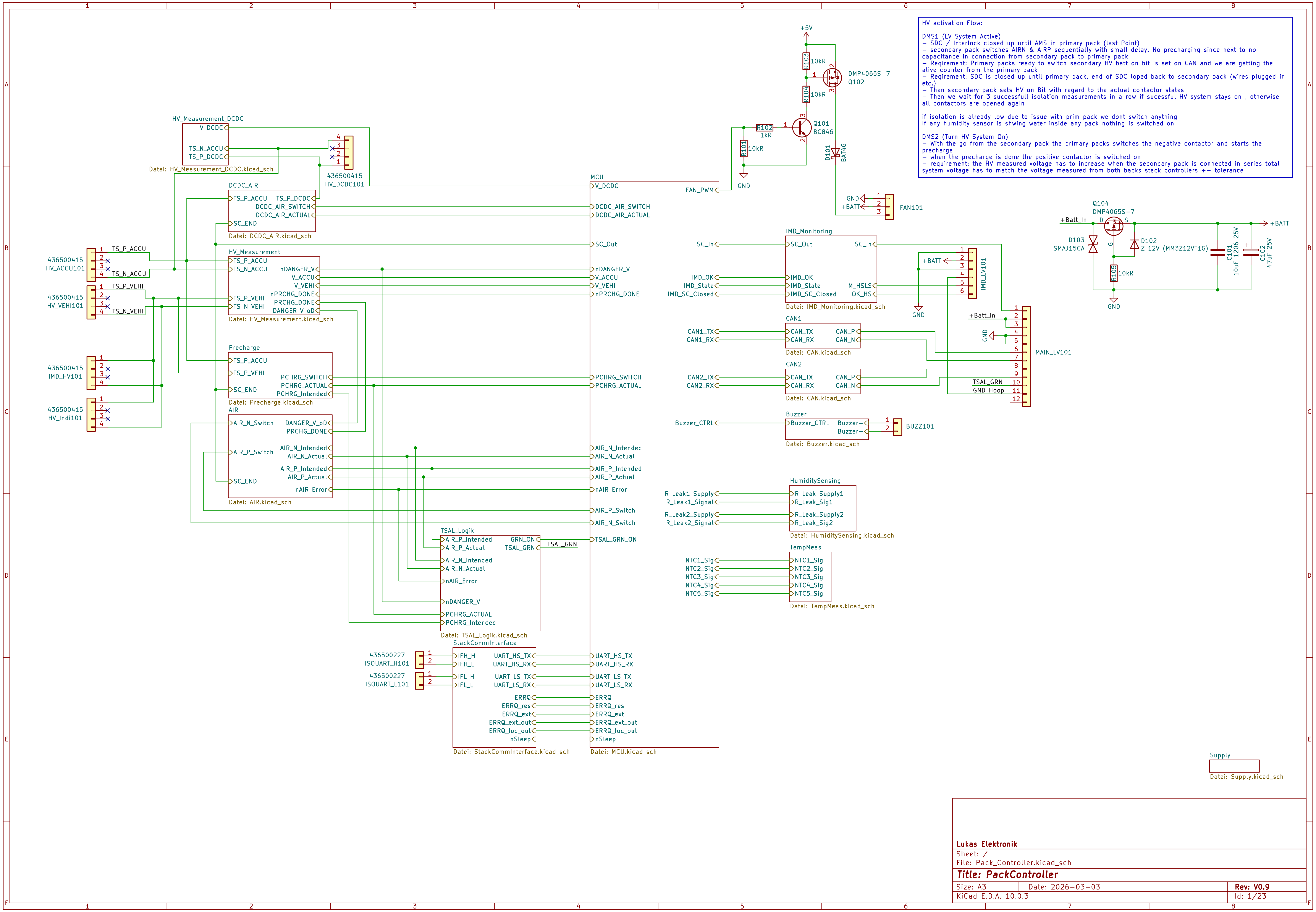


HV activation Flow:

DMS1 (LV System Active)
- SDC / Interlock closed up until AMS in primary pack (last Point)
- secondary pack switches AIRN & AIRP sequentially with small delay. No precharging since next to no capacitance in connection from secondary pack to primary pack
- Reqirement: Primary packs ready to switch secondary HV batt on bit is set on CAN and we are getting the alive counter from the primary pack
- Reqirement: SDC is closed up until primary pack, end of SDC loped back to secondary pack (wires plugged in etc.)
- Then secondary pack sets HV on Bit with regard to the actual contactor states
- Then we wait for 3 successfull isolation measurements in a row if sucessful HV system stays on , otherwise all contactors are opened again

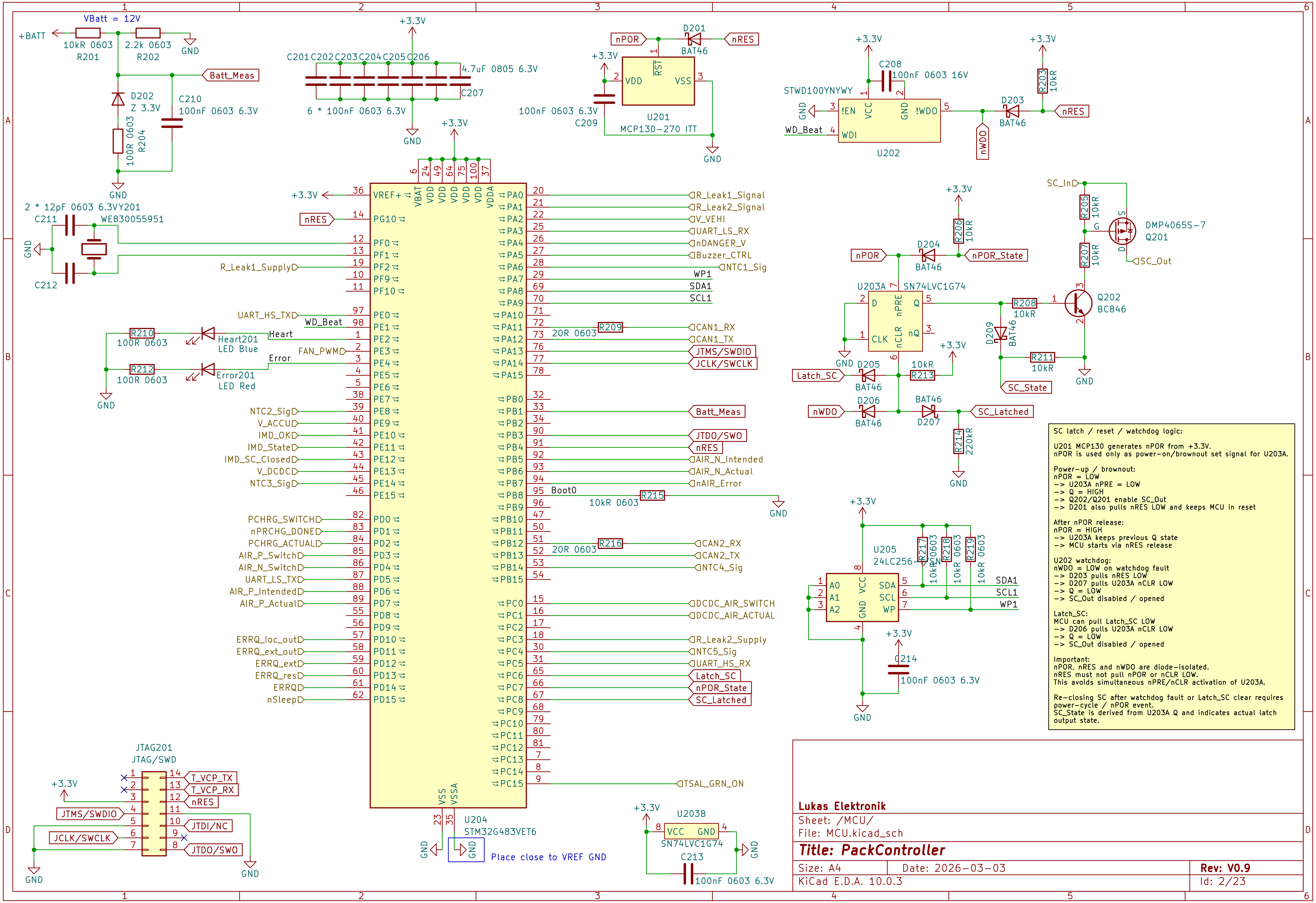
if isolation is already low due to issue with prim pack we dont switch anything
If any humidity sensor is shwing water inside any pack nothing is switched on

DMS2 (Turn HV System On)
- With the go from the secondary pack the primary packs switches the negative contactor and starts the precharge
- when the precharge is done the positive contactor is switched on
- requirement: the HV measured voltage has to increase when the secondary pack is connected in series total system voltage has to match the voltage measured from both backs stack controllers +- tolerance

Lukas Elektronik
Sheet: /
File: Pack_Controller.kicad_sch

Title: PackController

Size: A3 | Date: 2026-03-03 | **Rev: V0.9**
KiCad E.D.A. 10.0.3 | Id: 1/23



SC latch / reset / watchdog logic:

U201 MCP130 generates nPOR from +3.3V.
nPOR is used only as power-on/brownout set signal for U203A.

Power-up / brownout:
nPOR = LOW
-> U203A nPRE = LOW
-> Q = HIGH
-> Q202/Q201 enable SC_Out
-> D201 also pulls nRES LOW and keeps MCU in reset

After nPOR release:
nPOR = HIGH
-> U203A keeps previous Q state
-> MCU starts via nRES release

U202 watchdog:
nWDO = LOW on watchdog fault
-> D203 pulls nRES LOW
-> D207 pulls U203A nCLR LOW
-> Q = LOW
-> SC_Out disabled / opened

Latch_SC:
MCU can pull Latch_SC LOW
-> D206 pulls U203A nCLR LOW
-> Q = LOW
-> SC_Out disabled / opened

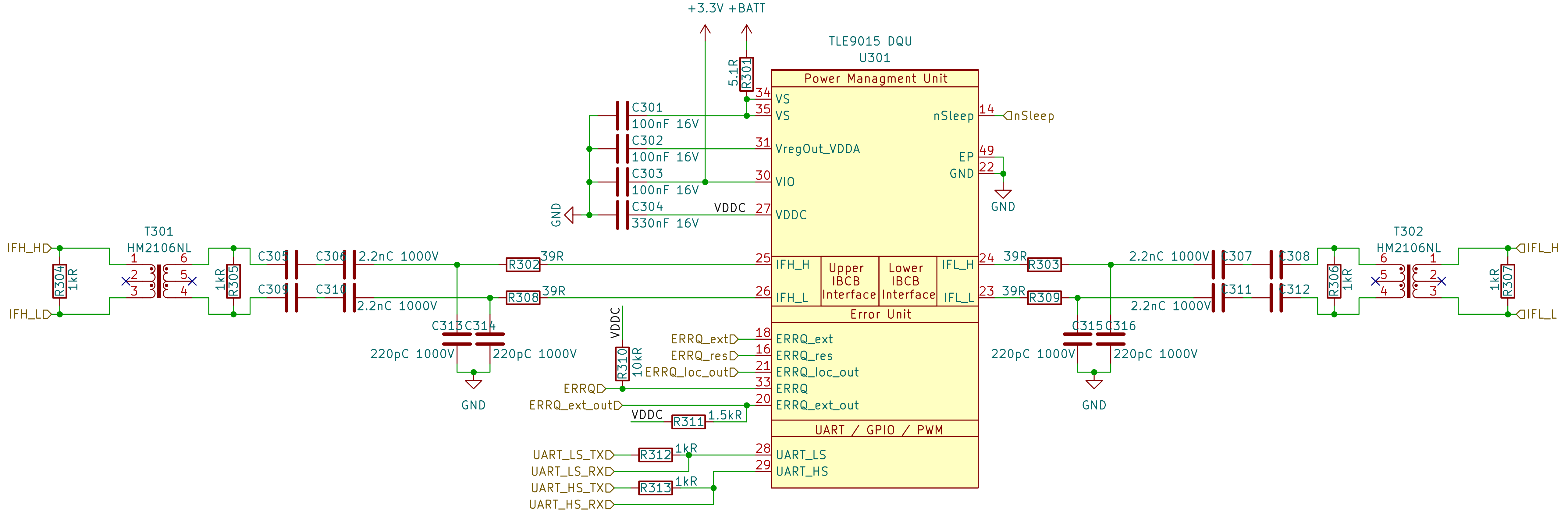
Important:
nPOR, nRES and nWDO are diode-isolated.
nRES must not pull nPOR or nCLR LOW.
This avoids simultaneous nPRE/nCLR activation of U203A.

Re-closing SC after watchdog fault or Latch_SC clear requires power-cycle / nPOR event.
SC_State is derived from U203A Q and indicates actual latch output state.

Lukas Elektronik
Sheet: /MCU/
File: MCU.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	Rev: V0.9
KiCad E.D.A. 10.0.3		Id: 2/23



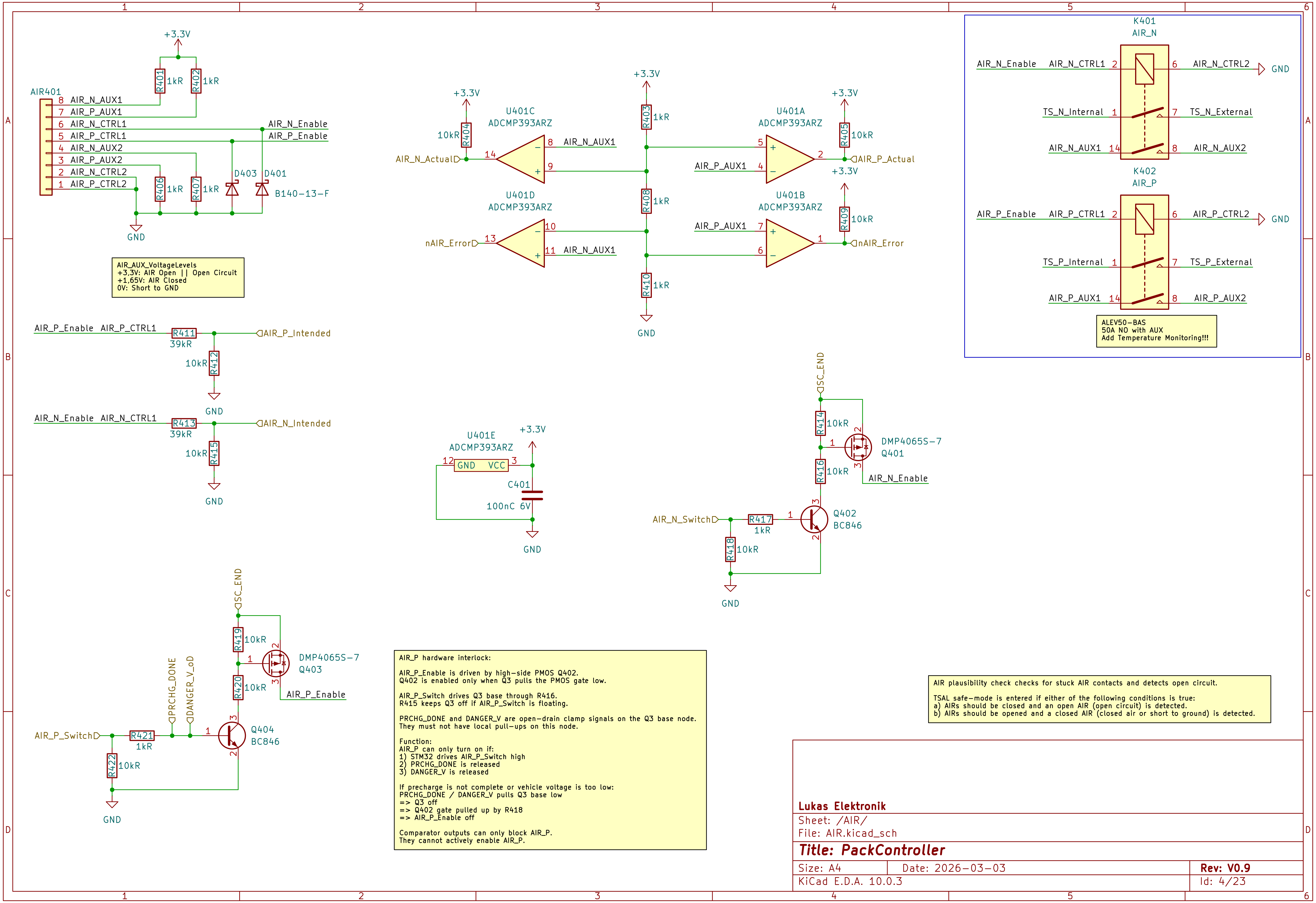
Lukas Elektronik

Sheet: /StackCommInterface/

File: StackCommInterface.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 3/23



AIR_P hardware interlock:

AIR_P_Enable is driven by high-side PMOS Q402.
Q402 is enabled only when Q3 pulls the PMOS gate low.

AIR_P_Switch drives Q3 base through R416.
R415 keeps Q3 off if AIR_P_Switch is floating.

PRCHG_DONE and DANGER_V are open-drain clamp signals on the Q3 base node.
They must not have local pull-ups on this node.

Function:
AIR_P can only turn on if:
1) STM32 drives AIR_P_Switch high
2) PRCHG_DONE is released
3) DANGER_V is released

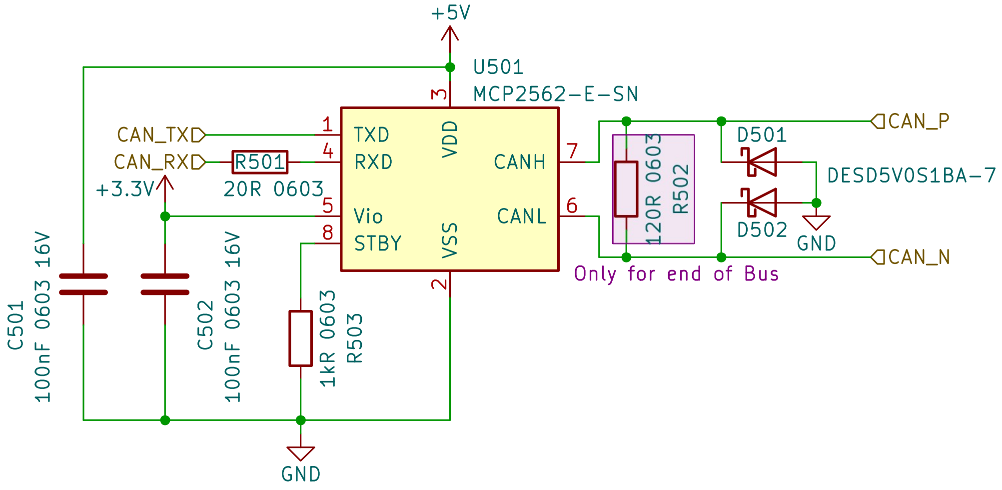
If precharge is not complete or vehicle voltage is too low:
PRCHG_DONE / DANGER_V pulls Q3 base low
=> Q3 off
=> Q402 gate pulled up by R418
=> AIR_P_Enable off

Comparator outputs can only block AIR_P.
They cannot actively enable AIR_P.

AIR plausibility check checks for stuck AIR contacts and detects open circuit.

TSAL safe-mode is entered if either of the following conditions is true:
a) AIRs should be closed and an open AIR (open circuit) is detected.
b) AIRs should be opened and a closed AIR (closed air or short to ground) is detected.

Lukas Elektronik
Sheet: /AIR/
File: AIR.kicad_sch
Title: PackController
Size: A4 Date: 2026-03-03 Rev: V0.9
KiCad E.D.A. 10.0.3 Id: 4/23



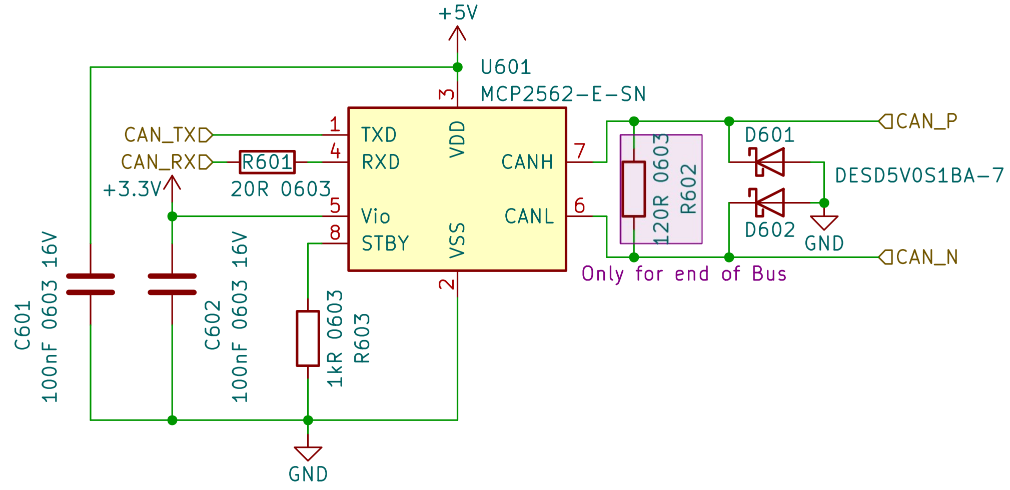
Lukas Elektronik

Sheet: /CAN1/

File: CAN.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 5/23

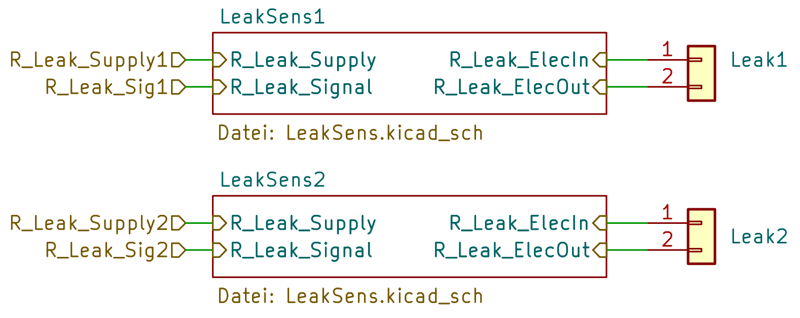


Lukas Elektronik

Sheet: /CAN2/
File: CAN.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 6/23

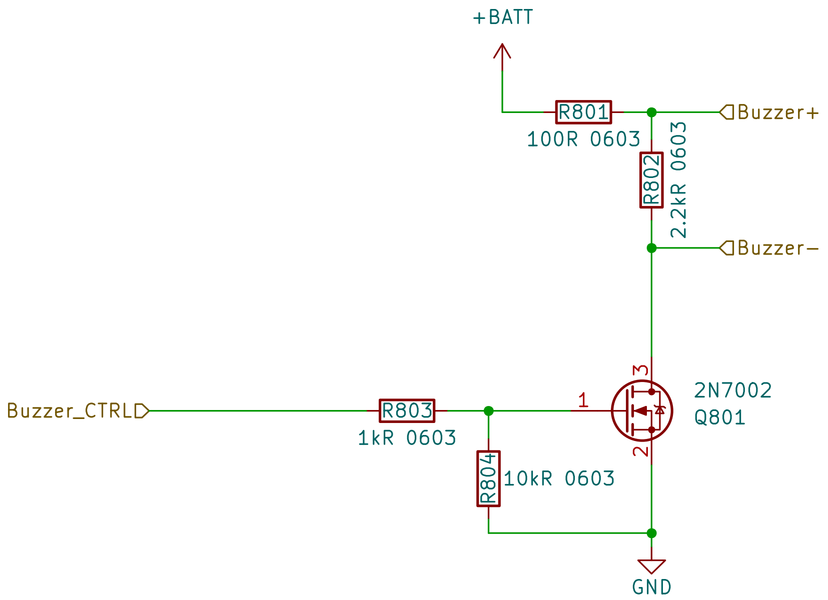


Lukas Elektronik

Sheet: /HumiditySensing/
File: HumiditySensing.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 7/23



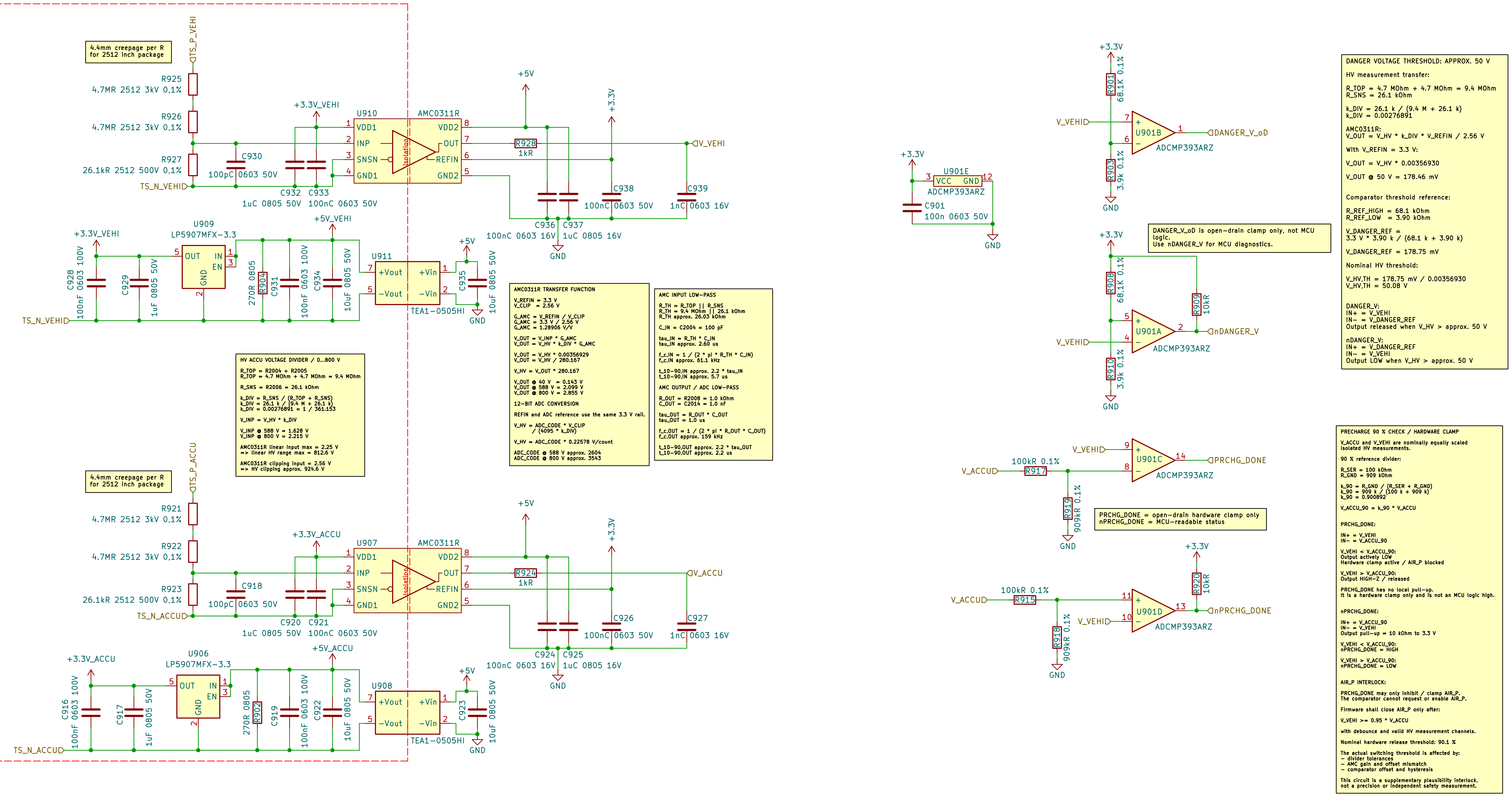
Passive piezo buzzer driver.
Target part: CPT−2305−90PM, externally driven, 12 Vp−p, 4 kHz, typ. 24 nF.

Drive with STM32 PWM:
f_PWM ≈ 4 kHz
duty ≈ 50 %

Q15 is a low−side NMOS switch.
R37 discharges the piezo during MOSFET off−time:
tau = R37 * C_PIEZO ≈ 2.2k * 24nF ≈ 53us.

R36 limits peak current and EMI.
No additional series coupling capacitor required.

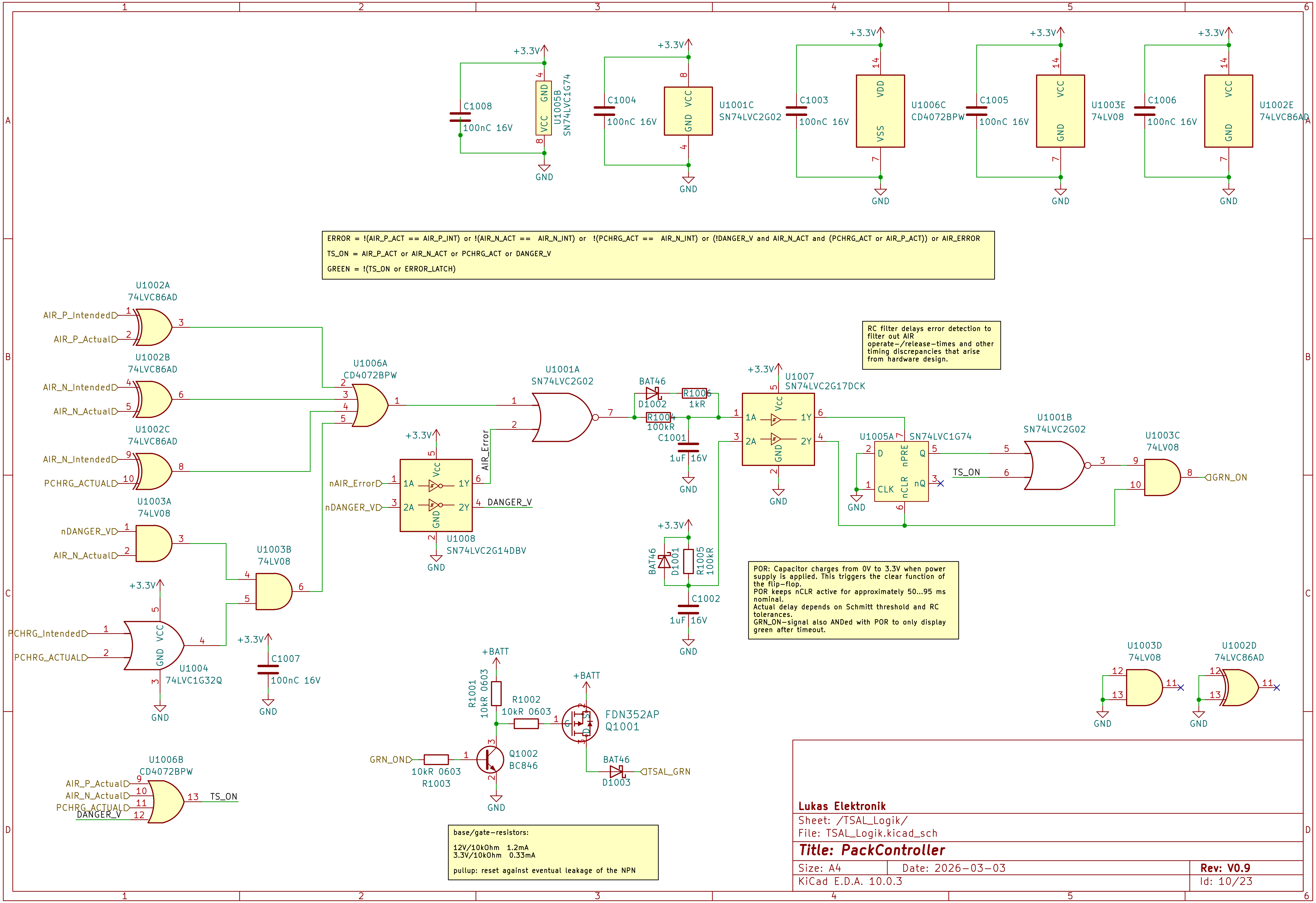
Lukas Elektronik		
Sheet: /Buzzer/		
File: Buzzer.kicad_sch		
Title: PackController		
Size: A4	Date: 2026−03−03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 8/23



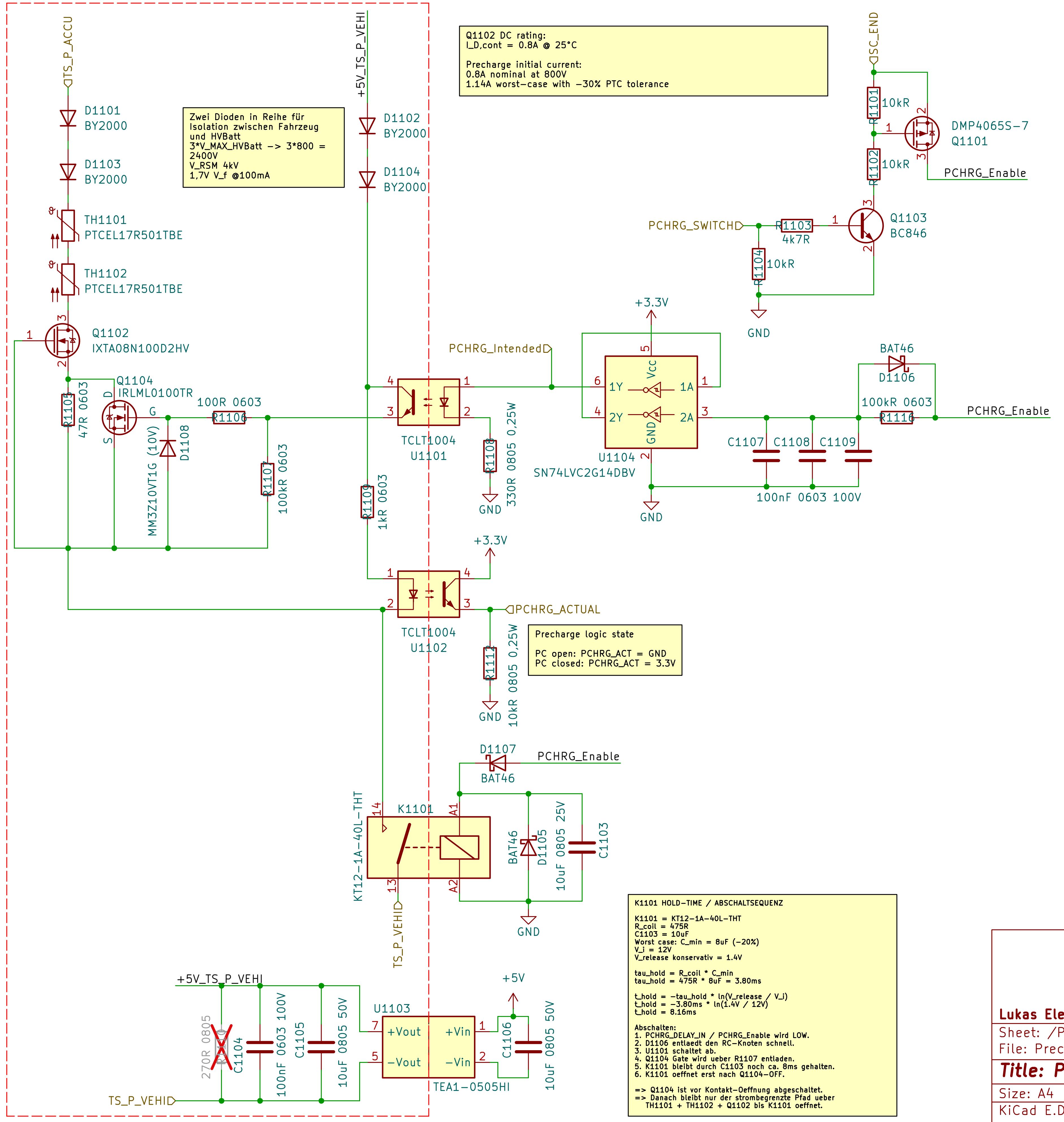
Lukas Elektronik
Sheet: /HV_Measurement/
File: HV_Measurement.kicad_sch

Title: PackController

Size: A3 | Date: 2026-03-03 | Rev: V0.9
KiCad E.D.A. 10.0.3 | Id: 9/23



ERROR = !(AIR_P_ACT == AIR_P_INT) or !(AIR_N_ACT == AIR_N_INT) or !(PCHRG_ACT == AIR_N_INT) or (!DANGER_V and AIR_N_ACT and (PCHRG_ACT or AIR_P_ACT)) or AIR_ERROR
TS_ON = AIR_P_ACT or AIR_N_ACT or PCHRG_ACT or DANGER_V
GREEN = !(TS_ON or ERROR_LATCH)
RC filter delays error detection to filter out AIR operate-/release-times and other timing discrepancies that arise from hardware design.
POR: Capacitor charges from 0V to 3.3V when power supply is applied. This triggers the clear function of the flip-flop.
POR keeps nCLR active for approximately 50...95 ms nominal.
Actual delay depends on Schmitt threshold and RC tolerances.
GRN_ON-signal also ANDed with POR to only display green after timeout.
base/gate-resistors:
12V/10kOhm 1.2mA
3.3V/10kOhm 0.33mA
pullup: reset against eventual leakage of the NPN
AIR_P_Intended
AIR_P_Actual
AIR_N_Intended
AIR_N_Actual
PCHRG_ACTUAL
nDANGER_V
PCHRG_Intended
nAIR_Error
AIR_Error
DANGER_V
TS_ON
GRN_ON
TSAL_GRN
+BATT
+3.3V
GND
U1002A 74LVC86AD
U1006A CD4072BPW
U1001A SN74LVC2G02
U1007 SN74LVC2G17DCK
U1005A SN74LVC1G74
U1001B SN74LVC2G02
U1003C 74LV08
U1008 SN74LVC2G14DBV
U1004 74LVC1G32Q
U1006B CD4072BPW
FDN352AP Q1001
Q1002 BC846
D1002 BAT46
D1001 BAT46
D1003 BAT46
R1006 1kR
R1004 100kR
R1005 100kR
C1001 1uF 16V
C1002 1uF 16V
R1001 10kR 0603
R1002 10kR 0603
R1003 10kR 0603
C1003 100nC 16V
C1004 100nC 16V
C1005 100nC 16V
C1006 100nC 16V
C1007 100nC 16V
C1008 100nC 16V
Lukas Elektronik
Sheet: /TSAL_Logik/
File: TSAL_Logik.kicad_sch
Title: PackController
Size: A4
Date: 2026-03-03
Rev: V0.9
KiCad E.D.A. 10.0.3
Id: 10/23



Zwei Dioden in Reihe für
Isolation zwischen Fahrzeug
und HVBatt
3*V_MAX_HVBatt -> 3*800 =
2400V
V_RSM 4kV
1,7V V_f @100mA

Q1102 DC rating:
I_D,cont = 0.8A @ 25°C

Precharge initial current:
0.8A nominal at 800V
1.14A worst-case with -30% PTC tolerance

Precharge logic state

PC open: PCHRG_ACT = GND
PC closed: PCHRG_ACT = 3.3V

K1101 HOLD-TIME / ABSCHALTSEQUENZ

K1101 = KT12-1A-40L-THT
R_coil = 475R
C1103 = 10uF
Worst case: C_min = 8uF (-20%)
V_i = 12V
V_release konservativ = 1.4V

tau_hold = R_coil * C_min
tau_hold = 475R * 8uF = 3.80ms

t_hold = -tau_hold * ln(V_release / V_i)
t_hold = -3.80ms * ln(1.4V / 12V)
t_hold = 8.16ms

Abschalten:
1. PCHRG_DELAY_IN / PCHRG_Enable wird LOW.
2. D1106 entlaedt den RC-Knoten schnell.
3. U1101 schaltet ab.
4. Q1104 Gate wird ueber R1107 entladen.
5. K1101 bleibt durch C1103 noch ca. 8ms gehalten.
6. K1101 oeffnet erst nach Q1104-OFF.

=> Q1104 ist vor Kontakt-Oeffnung abgeschaltet.
=> Danach bleibt nur der strombegrenzte Pfad ueber
TH1101 + TH1102 + Q1102 bis K1101 oeffnet.

Q1104 PRECHARGE GATE DRIVE / WORST CASE

Bestueckung:
Q1104 = IRLML0100TR
Q1102 = IXTA08N100D2HV
U1101, U1102 = TCLT1004
U1104 = SN74LVC2G14DBV
R1106 = 100R
R1107 = 100k
R1108 = 330R
R1109 = 1k
D1102 + D1104 = ca. 1.7V
D1105 = 10V Gate-Source Clamp

Architektur:
Q1104 brueckt nur Q1102.
TH1101 + TH1102 bleiben immer in Serie.
Q1104 ist daher nicht der Hauptstrom-Bypass wie bei DCDC_AIR,
sondern reduziert nur den Spannungsabfall / Verlust an Q1102
nachdem der Precharge-Pfad aktiv ist.

Annahmen Gate-Drive:
+5V_TS_P_VEHI liegt vor D1102/D1104
V_C,U1101 = 5.0V - 1.7V = 3.3V
VCEsat,U1101,max = 0.3V
=> V_GS,Q1104,final,worst = 3.3V - 0.3V = 3.0V

U1101 Eingang:
IF_U1101,min = (3.3V - 1.6V) / 330R = 5.15mA
IF_U1101,typ = (3.3V - 1.25V) / 330R = 6.21mA
IF_limit = 60mA
=> U1101 LED-Strom OK

R1108 Verlustleistung:
P_R1108,typ = (3.3V - 1.25V)^2 / 330R = 12.7mW
=> R1108 thermisch unkritisch

U1101 Ausgangsreserve:
konservativ mit CTR_min = 56%:
IC_U1101,min = 0.56 * 5.15mA = 2.88mA

U1101 statischer Strombedarf bei V_GS = 3.0V:
I_R1107 = 3.0V / 100k = 0.03mA
I_U1101_req,static ~= 0.03mA

=> U1101 Reserve statisch > Faktor 90
=> Gate-Endspannung Q1104 worst case ca. 3.0V

Hinweis:
Es gibt keine grosse Gate-Timing-Kapazitaet mehr.
Das Timing wird durch LV-RC + U1104 bestimmt.

Q1104 EINSCHALTEN / LV-DELAY

Bestueckung:
R1116 = 100k
C1107 = 100nF
C1108 = 100nF
C1109 = 100nF
C_sum = 300nF
U1104 = SN74LVC2G14DBV, zwei Inverter in Serie

Wichtig:
U1104 darf nur mit einem 3.3V-kompatiblen Signal betrieben werden.
PCHRG_DELAY_IN = 3.3V-kompatibles, begrenztes PCHRG_Enable-Signal.

tau_ON = R1116 * C_sum
tau_ON = 100k * 300nF = 30ms

Schmitt-Schwelle angenommen:
V_T+ = 1.6...2.3V bei VCC = 3.3V

t_ON = -tau_ON * ln(1 - V_T+ / 3.3V)

Fuer V_T+ = 1.6V:
t_ON = -30ms * ln(1 - 1.6V / 3.3V) = 20ms

Fuer V_T+ = 2.3V:
t_ON = -30ms * ln(1 - 2.3V / 3.3V) = 36ms

=> U1101 / Q1104 wird ca. 20...36ms nach
PCHRG_DELAY_IN eingeschaltet.

Gate-Ladung Q1104 ist gegenueber 20...36ms vernachlaessigbar.
=> Q1104-ON wird durch R1116/C1107/C1108/C1109 bestimmt,
nicht durch das MOSFET-Gate.

Q1104 AUSSCHALTEN

Bei PCHRG_DELAY_IN = LOW:
D1106 entlaedt den RC-Knoten schnell.
U1104 schaltet zurueck.
U1101 LED geht aus.
Q1104 Gate entlaedt ueber R1107.

R1107 = 100k
Keine externe Gate-Timing-Kapazitaet vorhanden.

=> Q1104-OFF ist deutlich schneller als K1101-Release.
=> Q1104 wird vor dem mechanischen Oeffnen von K1101 abgeschaltet.

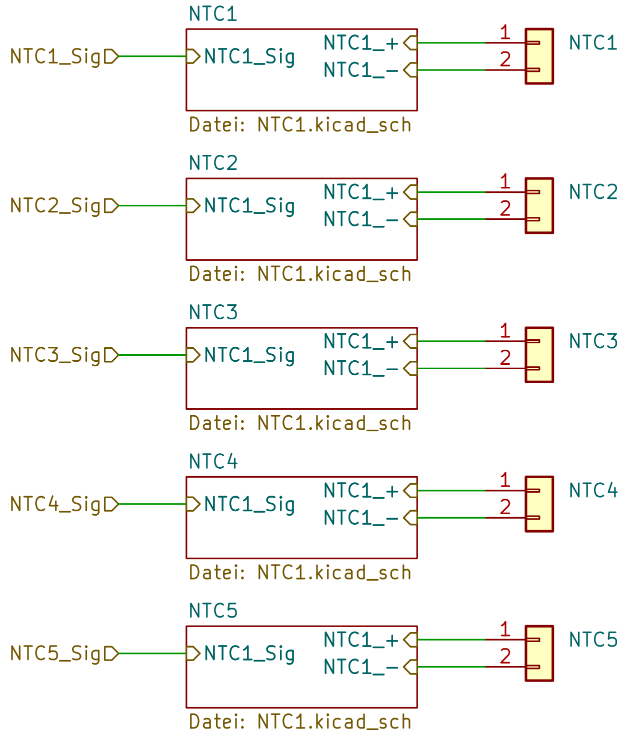
Wichtig:
Nach Q1104-OFF bleibt der Precharge-Pfad nicht hart aus,
sondern laeuft bis K1101 oeffnet weiterhin ueber:
TH1101 + TH1102 + Q1102

Das ist beabsichtigt:
Q1104 ist nur der Q1102-Bypass, nicht der alleinige Precharge-Schalter.

Lukas Elektronik
Sheet: /Precharge/
File: Precharge.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	Rev: V0.9
KiCad E.D.A. 10.0.3		Id: 11/23

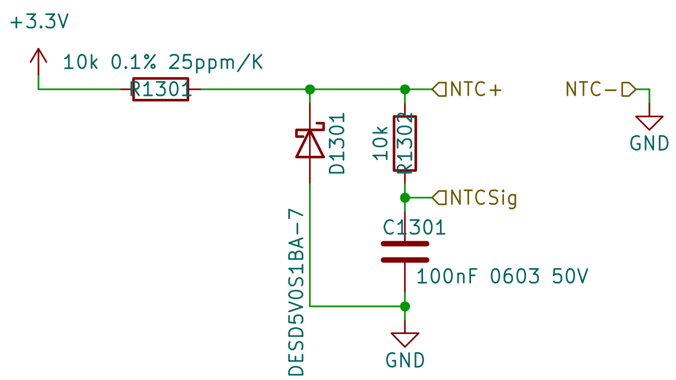


Lukas Elektronik

Sheet: /TempMeas/
File: TempMeas.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 12/23



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

R_NTC = R_PULLUP * x / (1 - x)
x = ADC_raw / ADC_max

R58/C49 form ADC input filter:
tau = 10k * 100nF = 1ms, fc ≈ 159Hz.

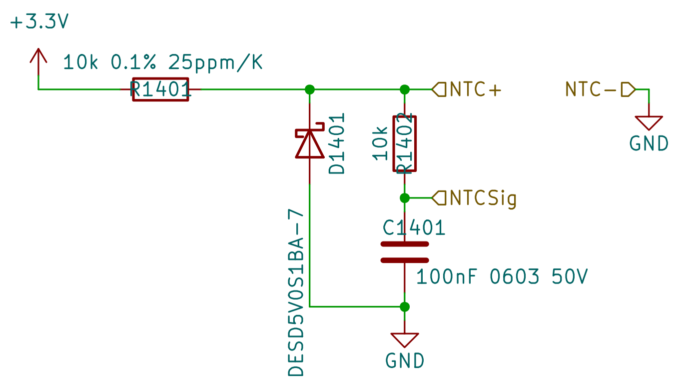
Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

Lukas Elektronik

Sheet: /TempMeas/NTC1/
File: NTC1.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 13/23



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

R_NTC = R_PULLUP * x / (1 − x)
x = ADC_raw / ADC_max

R58/C49 form ADC input filter:
tau = 10k * 100nF = 1ms, fc ≈ 159Hz.

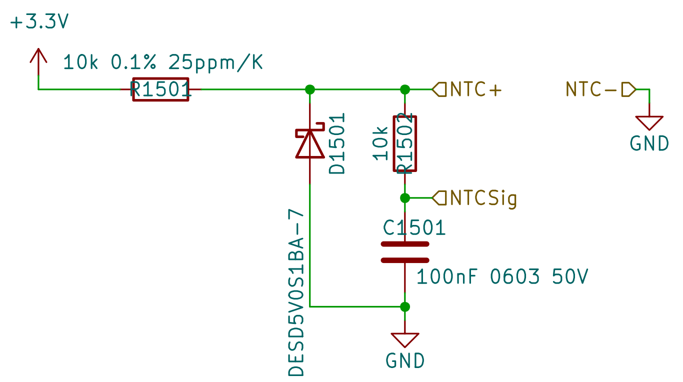
Open NTC −> ADC near full scale.
Short to GND −> ADC near 0.

Lukas Elektronik

Sheet: /TempMeas/NTC2/
File: NTC1.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 14/23



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

R_NTC = R_PULLUP * x / (1 - x)
x = ADC_raw / ADC_max

R58/C49 form ADC input filter:
tau = 10k * 100nF = 1ms, fc ≈ 159Hz.

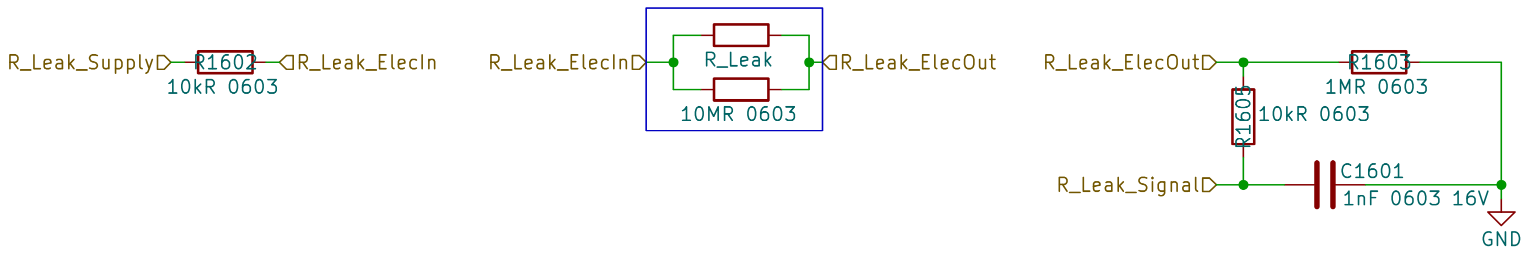
Open NTC --> ADC near full scale.
Short to GND --> ADC near 0.

Lukas Elektronik

Sheet: /TempMeas/NTC3/
File: NTC1.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 15/23



Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open−wire detection.

R_SENSOR = R_LEAK || R_EOL

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi−Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
x = ADC_CODE / ADC_MAX

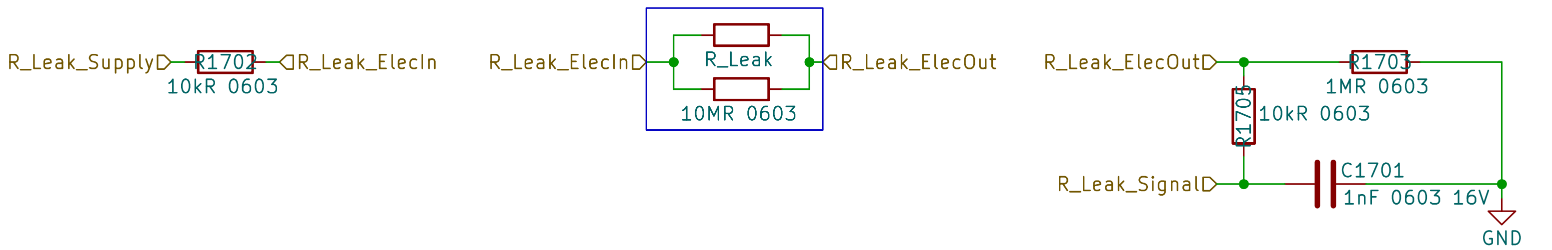
R_SENSOR = R_REF × (1/x − 1)− R_SER_SUPPLY

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC ≈ 0
dry connected: R_SENSOR ≈ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k

Lukas Elektronik		
Sheet: /HumiditySensing/LeakSens1/		
File: LeakSens.kicad_sch		
Title: PackController		
Size: A4	Date: 2026−03−03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 16/23



Leakage sensor:
Flex electrodes form R_LEAK. R_EOL = 10M is mounted on the flex sensor,
parallel to the electrodes, for open–wire detection.

R_SENSOR = R_LEAK || R_EOL

Measurement:
Pulse R_Leak_Supply HIGH, wait 10...50ms, sample ADC, then set supply LOW/Hi–Z.
Suggested scan rate: 1Hz.

ADC conversion, assuming ADC reference = VDD and R_Leak_Supply = VDD:
x = ADC_CODE / ADC_MAX

R_SENSOR = R_REF × (1/x − 1)− R_SER_SUPPLY

Nominal:
R_SER_SUPPLY = 10k
R_REF = 1M
R_ADC_SER = 10k
C_ADC = 1nF
R_EOL = 10M on flex

Expected:
disconnected/open: ADC ≈ 0
dry connected: R_SENSOR ≈ 10M
wet/leakage: R_SENSOR decreases

Initial thresholds, validate with real coolant:
warning: R_SENSOR < 3M
leak: R_SENSOR < 1M
severe leak: R_SENSOR < 300k

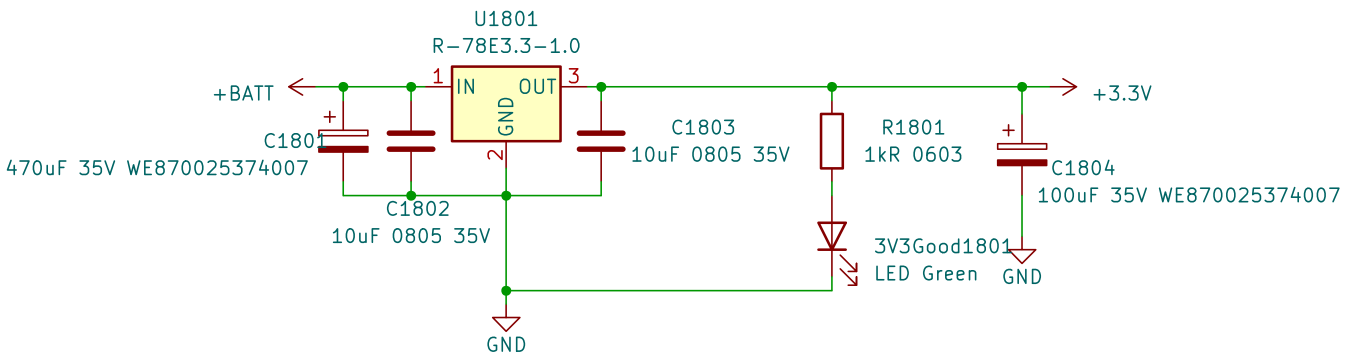
Lukas Elektronik		
Sheet: /HumiditySensing/LeakSens2/		
File: LeakSens.kicad_sch		
Title: PackController		
Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 17/23

+3V3 Supply worst-case estimate:

Main loads:
STM32G483 active incl. peripherals: ~50 mA
ADuM3190 VDD1, 2x: ~4 mA
24LC256 EEPROM write: ~3 mA
MCP2562 VIO, 2x worst-case: ~1 mA
ADCMP393 comparators + logic: <1 mA
Pullups, dividers, GPIO drivers, LEDs: ~20...30 mA

Estimated worst-case load: ~80...90 mA
Selected supply: RECOM 3.3 V / 1 A DC/DC

Design margin:
1 A / 90 mA ≈ 11x

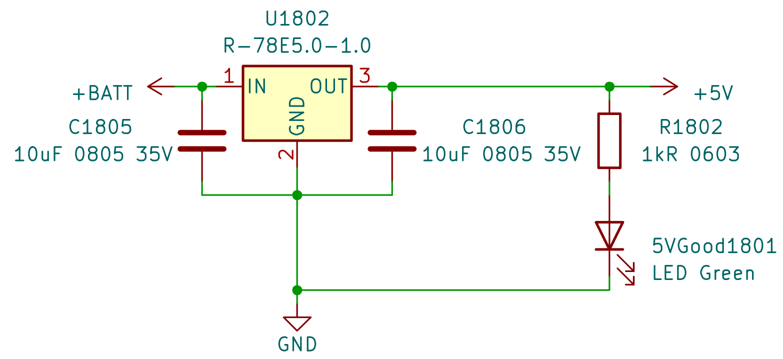


+5V Supply worst-case estimate:

Main loads:
TEA1-0505HI isolated DC/DC no-load, 5x: ~120 mA
TEA1 loads incl. ADuM3190 isolated sides: ~25...30 mA
MCP2562 CAN transceivers, 2x dominant: ~140 mA
Misc. reserve / startup / tolerance: ~50 mA

Estimated worst-case load: ~330...400 mA
Selected supply: RECOM 5 V / 1 A DC/DC

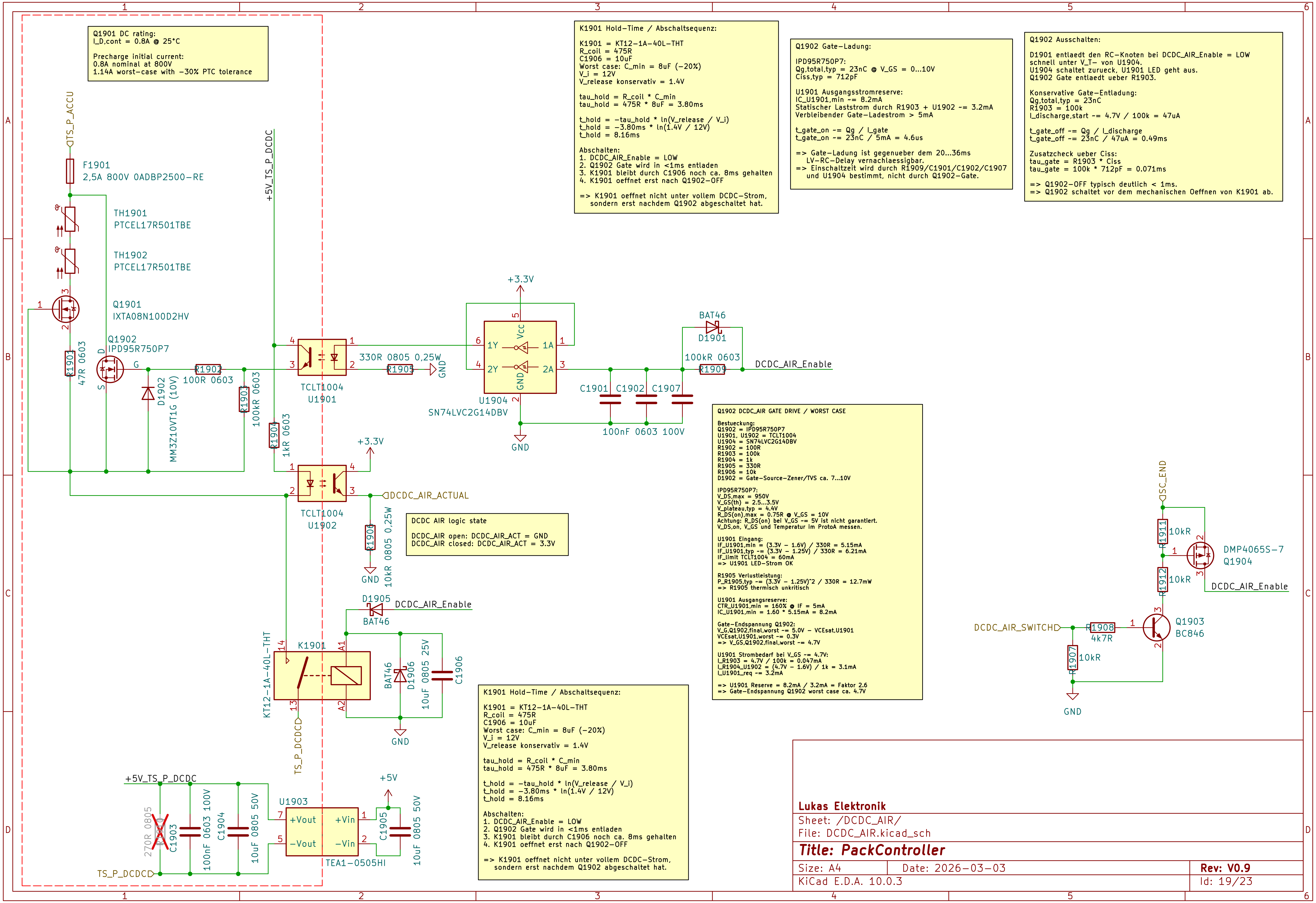
Design margin:
Design margin: 1A / 0.4A ≈ 2.5x



Lukas Elektronik
Sheet: /Supply/
File: Supply.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 18/23



Q1901 DC rating:
I_D,cont = 0.8A @ 25°C

Precharge initial current:
0.8A nominal at 800V
1.14A worst-case with -30% PTC tolerance

K1901 Hold-Time / Abschaltsequenz:

K1901 = KT12-1A-40L-THT
R_coil = 475R
C1906 = 10uF
Worst case: C_min = 8uF (-20%)
V_i = 12V
V_release konservativ = 1.4V

tau_hold = R_coil * C_min
tau_hold = 475R * 8uF = 3.80ms

t_hold = -tau_hold * ln(V_release / V_i)
t_hold = -3.80ms * ln(1.4V / 12V)
t_hold = 8.16ms

Abschalten:
1. DCDC_AIR_Enable = LOW
2. Q1902 Gate wird in <1ms entladen
3. K1901 bleibt durch C1906 noch ca. 8ms gehalten
4. K1901 oeffnet erst nach Q1902-OFF

=> K1901 oeffnet nicht unter vollem DCDC-Strom,
sondern erst nachdem Q1902 abgeschaltet hat.

Q1902 Gate-Ladung:

IPD95R750P7:
Qg,total,typ = 23nC @ V_GS = 0...10V
Ciss,typ = 712pF

U1901 Ausgangsstromreserve:
IC_U1901,min ~= 8.2mA
Statischer Laststrom durch R1903 + U1902 ~= 3.2mA
Verbleibender Gate-Ladestrom > 5mA

t_gate_on ~= Qg / I_gate
t_gate_on ~= 23nC / 5mA = 4.6us

=> Gate-Ladung ist gegenueber dem 20...36ms
LV-RC-Delay vernachlaessigbar.
=> Einschaltzeit wird durch R1909/C1901/C1902/C1907
und U1904 bestimmt, nicht durch Q1902-Gate.

Q1902 Ausschalten:

D1901 entlaedt den RC-Knoten bei DCDC_AIR_Enable = LOW
schnell unter V_T- von U1904.
U1904 schaltet zurueck, U1901 LED geht aus.
Q1902 Gate entlaedt ueber R1903.

Konservative Gate-Entladung:
Qg,total,typ = 23nC
R1903 = 100k
I_discharge,start ~= 4.7V / 100k = 47uA

t_gate_off ~= Qg / I_discharge
t_gate_off ~= 23nC / 47uA = 0.49ms

Zusatzcheck ueber Ciss:
tau_gate = R1903 * Ciss
tau_gate = 100k * 712pF = 0.071ms

=> Q1902-OFF typisch deutlich < 1ms.
=> Q1902 schaltet vor dem mechanischen Oeffnen von K1901 ab.

Q1902 DCDC_AIR GATE DRIVE / WORST CASE

Bestueckung:
Q1902 = IPD95R750P7
U1901, U1902 = TCLT1004
U1904 = SN74LVC2G14DBV
R1902 = 100R
R1903 = 100k
R1904 = 1k
R1905 = 330R
R1906 = 10k
D1902 = Gate-Source-Zener/TVS ca. 7...10V

IPD95R750P7:
V_DS,max = 950V
V_GS(th) = 2.5...3.5V
V_plateau,typ = 4.4V
R_DS(on),max = 0.75R @ V_GS = 10V
Achtung: R_DS(on) bei V_GS ~= 5V ist nicht garantiert.
V_DS,on, V_GS und Temperatur im ProtoA messen.

U1901 Eingang:
IF_U1901,min = (3.3V - 1.6V) / 330R = 5.15mA
IF_U1901,typ ~= (3.3V - 1.25V) / 330R = 6.21mA
IF_limit TCLT1004 = 60mA
=> U1901 LED-Strom OK

R1905 Verlustleistung:
P_R1905,typ ~= (3.3V - 1.25V)^2 / 330R = 12.7mW
=> R1905 thermisch unkritisch

U1901 Ausgangsreserve:
CTR_U1901,min = 160% @ IF = 5mA
IC_U1901,min = 1.60 * 5.15mA = 8.2mA

Gate-Endspannung Q1902:
V_G,Q1902,final,worst ~= 5.0V - VCEsat,U1901
VCEsat,U1901,worst ~= 0.3V
=> V_GS,Q1902,final,worst ~= 4.7V

U1901 Strombedarf bei V_GS ~= 4.7V:
I_R1903 = 4.7V / 100k = 0.047mA
I_R1904_U1902 = (4.7V - 1.6V) / 1k = 3.1mA
I_U1901_req ~= 3.2mA

=> U1901 Reserve = 8.2mA / 3.2mA = Faktor 2.6
=> Gate-Endspannung Q1902 worst case ca. 4.7V

DCDC AIR logic state

DCDC_AIR open: DCDC_AIR_ACT = GND
DCDC_AIR closed: DCDC_AIR_ACT = 3.3V

K1901 Hold-Time / Abschaltsequenz:

K1901 = KT12-1A-40L-THT
R_coil = 475R
C1906 = 10uF
Worst case: C_min = 8uF (-20%)
V_i = 12V
V_release konservativ = 1.4V

tau_hold = R_coil * C_min
tau_hold = 475R * 8uF = 3.80ms

t_hold = -tau_hold * ln(V_release / V_i)
t_hold = -3.80ms * ln(1.4V / 12V)
t_hold = 8.16ms

Abschalten:
1. DCDC_AIR_Enable = LOW
2. Q1902 Gate wird in <1ms entladen
3. K1901 bleibt durch C1906 noch ca. 8ms gehalten
4. K1901 oeffnet erst nach Q1902-OFF

=> K1901 oeffnet nicht unter vollem DCDC-Strom,
sondern erst nachdem Q1902 abgeschaltet hat.

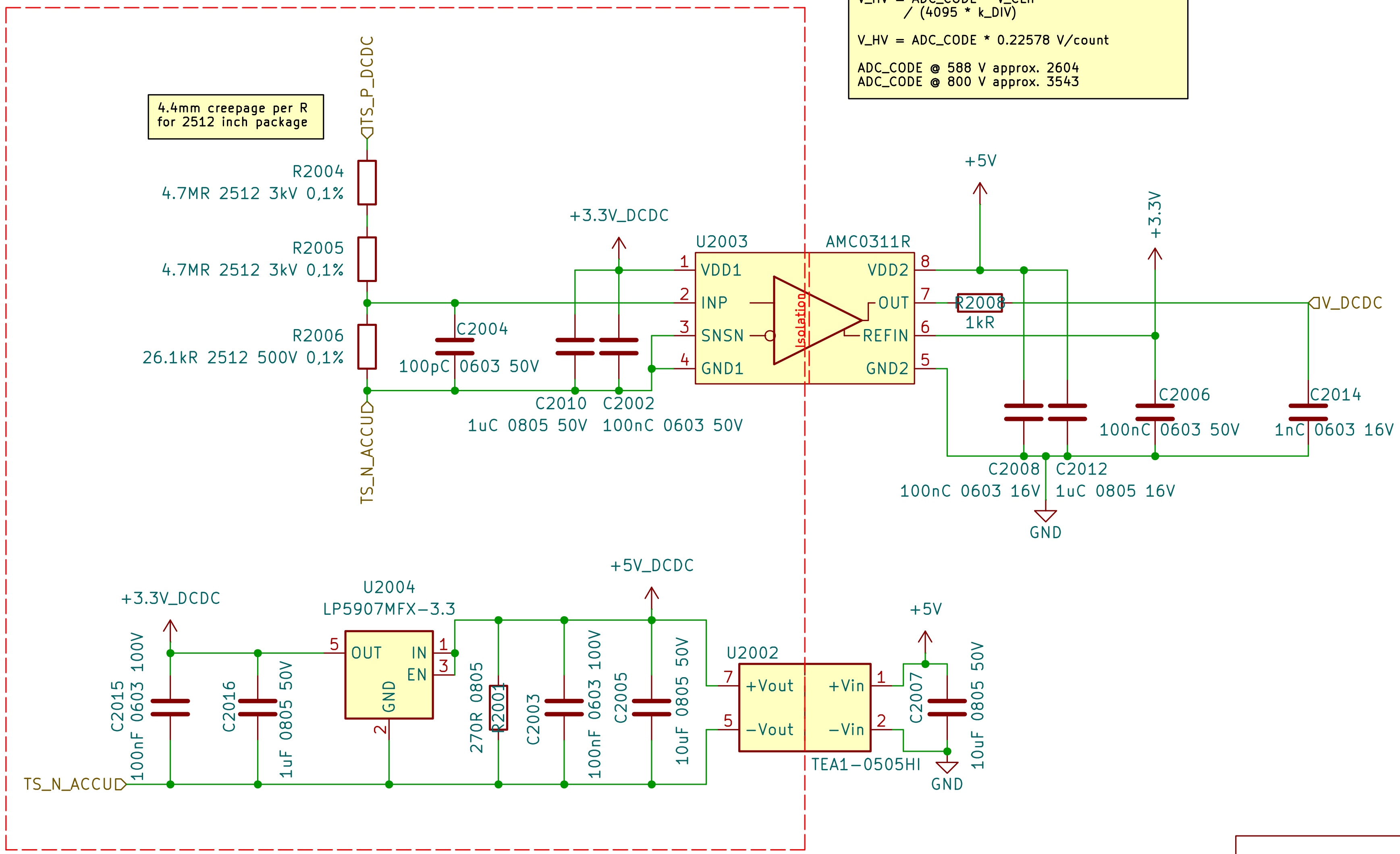
Lukas Elektronik
Sheet: /DCDC_AIR/
File: DCDC_AIR.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	Rev: V0.9
KiCad E.D.A. 10.0.3		Id: 19/23

```
HV ACCU VOLTAGE DIVIDER / 0...800 V

R_TOP = R2004 + R2005
R_TOP = 4.7 MOhm + 4.7 MOhm = 9.4
MOhm

R_SNS = R2006 = 26.1 kOhm

k_DIV = R_SNS / (R_TOP + R_SNS)
k_DIV = 26.1 k / (9.4 M + 26.1 k)
k_DIV = 0.00276891 = 1 / 361.153

V_INP = V_HV * k_DIV

V_INP @ 588 V = 1.628 V
V_INP @ 800 V = 2.215 V

AMC0311R linear input max = 2.25 V
=> linear HV range max = 812.6 V

AMC0311R clipping input = 2.56 V
=> HV clipping approx. 924.6 V
```

```
AMC0311R TRANSFER FUNCTION

V_REFIN = 3.3 V
V_CLIP  = 2.56 V

G_AMC = V_REFIN / V_CLIP
G_AMC = 3.3 V / 2.56 V
G_AMC = 1.28906 V/V

V_OUT = V_INP * G_AMC
V_OUT = V_HV * k_DIV * G_AMC

V_OUT = V_HV * 0.00356929
V_OUT = V_HV / 280.167

V_HV = V_OUT * 280.167

V_OUT @ 40 V  = 0.143 V
V_OUT @ 588 V = 2.099 V
V_OUT @ 800 V = 2.855 V

12-BIT ADC CONVERSION

REFIN and ADC reference use the same 3.3 V
rail.

V_HV = ADC_CODE * V_CLIP
       / (4095 * k_DIV)

V_HV = ADC_CODE * 0.22578 V/count

ADC_CODE @ 588 V approx. 2604
ADC_CODE @ 800 V approx. 3543
```



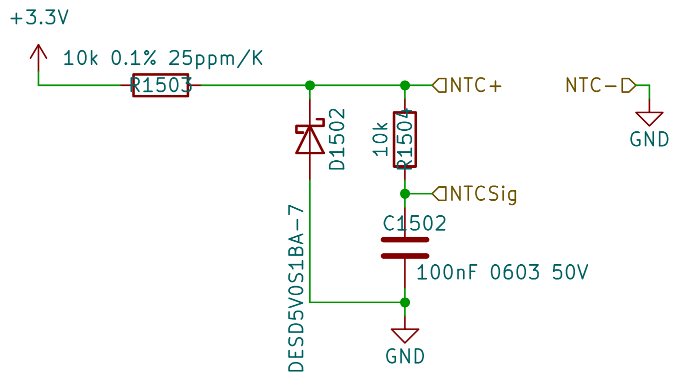

```
AMC INPUT LOW-PASS

R_TH = R_TOP || R_SNS
R_TH = 9.4 MOhm || 26.1 kOhm
R_TH approx. 26.03 kOhm

C_IN = C2004 = 100 pF

tau_IN = R_TH * C_IN
tau_IN approx. 2.60 us

f_c,IN = 1 / (2 * pi * R_TH * C_IN)
f_c,IN approx. 61.1 kHz

t_10-90,IN approx. 2.2 * tau_IN
t_10-90,IN approx. 5.7 us

AMC OUTPUT / ADC LOW-PASS

R_OUT = R2008 = 1.0 kOhm
C_OUT = C2014 = 1.0 nF

tau_OUT = R_OUT * C_OUT
tau_OUT = 1.0 us

f_c,OUT = 1 / (2 * pi * R_OUT * C_OUT)
f_c,OUT approx. 159 kHz

t_10-90,OUT approx. 2.2 * tau_OUT
t_10-90,OUT approx. 2.2 us
```

Lukas Elektronik

Sheet: /HV_Measurement_DCDC/
File: HV_Measurement_DCDC.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 20/23




```
External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

R_NTC = R_PULLUP * x / (1 - x)
x = ADC_raw / ADC_max

R58/C49 form ADC input filter:
tau = 10k * 100nF = 1ms, fc ≈ 159Hz.

Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.
```

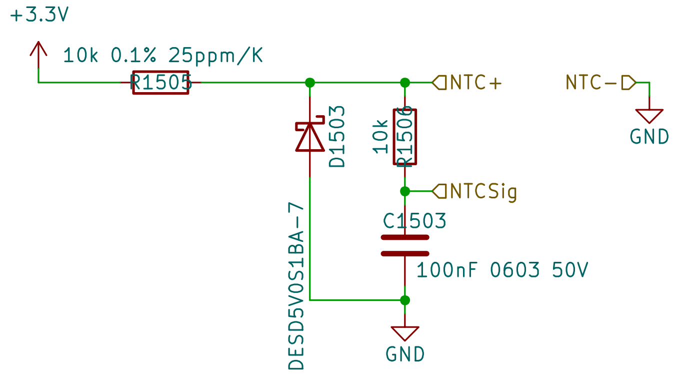
Lukas Elektronik

Sheet: /TempMeas/NTC4/

File: NTC1.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 21/23



External NTC input:
10k NTC to GND, 10k 0.1% pullup to 3V3.
ADC is ratiometric to 3V3.

R_NTC = R_PULLUP * x / (1 - x)
x = ADC_raw / ADC_max

R58/C49 form ADC input filter:
tau = 10k * 100nF = 1ms, fc ≈ 159Hz.

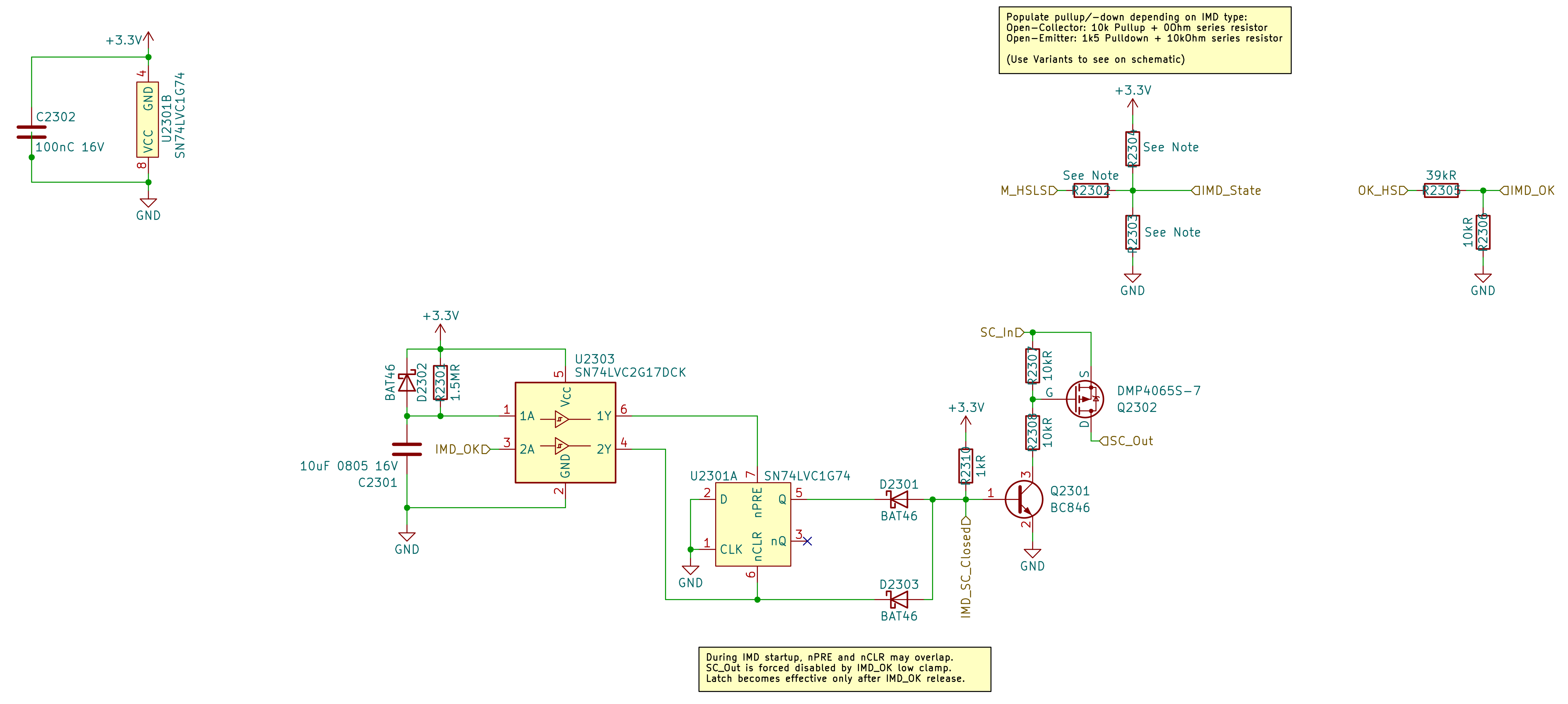
Open NTC -> ADC near full scale.
Short to GND -> ADC near 0.

Lukas Elektronik

Sheet: /TempMeas/NTC5/
File: NTC1.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 22/23



Lukas Elektronik

Sheet: /IMD_Monitoring/

File: IMD_Monitoring.kicad_sch

Title: PackController

Size: A4	Date: 2026-03-03	**Rev: V0.9**
KiCad E.D.A. 10.0.3		Id: 23/23